

Constructive constraints: On the role of chance and complexity in artistic creativity

Possibility Studies & Society

2023, Vol. 1(3) 311–323

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DOI: 10.1177/27538699231193539

journals.sagepub.com/home/pst

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Abstract

Creativity is sometimes understood as the individual, mental creation of novelty which is then imposed as form onto a material substrate. On this view, constraints would feature only as obstacles in the creative process. In contrast, we support the view that creativity emerges from the interactions between human agents and the material, cultural, and social constraints that comprise a particular situation. In this account, constraints play a key enabling role, scaffolding the space of possibilities in which improvisation and chance give rise to novelty. In this paper, we review experimental work on interpersonal synergies between jazz pianists, which provides a scientific framework from which to approach the complex relationship between chance, constraints, and creativity. We also explore two case studies from the science of complexity and the creative arts in order to substantiate this notion of creative constraints: Anthony Braxton's musical innovation at the intersection between composition and improvisation, and Marcel Duchamp's radical interventions in the fields of sculpture and conceptual art.

Keywords

Anthony Braxton, complexity, constraints, creativity, jazz, Marcel Duchamp

Introduction¹

A commonsense view might assume that creativity flourishes in maximal freedom and the absence of constraints. One way this manifests in everyday classroom life is when vague assignments are accompanied with the cheerful addition “you can be really creative on this one!” This apparent freedom from constraints often does not lead to us feeling inspired, but rather

somewhat lost, with no clear direction or structure for our task. This is because the whole idea of creativity as a force emerging freely from itself or from some spark of genius, and of constraints as a mere obstacle that has to be

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overcome or removed to make way for creative expression is fundamentally mistaken. In this paper, we briefly characterize this mistaken view of the relationship between constraints and creativity, and then offer an alternative account on which constraints play a crucial role in enabling and shaping creativity, rather than hindering or blocking it.

Besides an appreciation of constraints as conducive to creativity, this alternative view also locates the origin of novelty not within the mind of some genius creator, but understands it as emerging from the complex dynamics of systems comprising the interactions between heterogeneous material structures and processes as well as the human and nonhuman agents that participate in them. Crucially, this view of creativity as constrained by material structures and processes is at odds with a common view of creativity within the cognitive sciences (e.g. Boden, 2004). Instead, we will present a view of creativity as emerging from the constraints of a complex dynamic system. The science of complexity allows us to identify and understand phenomena that occur on different scales, across different material substrates, and within a variety of cultural contexts. From this perspective, we understand creative achievements as involving neither inexplicable momentary strokes of genius nor abstract representations that are realized according to a strict plan. Instead, they emerge from specific kinds of conditions within complex systems.

We illustrate how the introduction of chance plays a crucial role in the emergence of novelty in the creative arts and in musical improvisation. Two case studies help us develop a more nuanced account of how constraints interact with creativity in the arts: Anthony Braxton's music shows how chance may be employed as an additional constraint that mediates the complex relationship between composition and improvisation, while Marcel Duchamp's radical intervention and introduction of chance into the way we understand sculpture highlights the multiple dependencies between material, cultural, and human constraints. What both Braxton and

Duchamp demonstrate in their artistic works is how chance may add additional constraints that act as a source of creativity in artistic contexts. We believe that the systematic study of these conditions can help us understand better how creativity operates in the wild.

Unmaking the hylomorphic account of creative production

A handful of authors have recently pointed out the importance of the relationship between constraints and creativity. For example, Damadzic et al. (2022) provide a meta-analysis of literature on constraints and creativity and find that most studies support the claim that constraints positively impact creativity. Importantly, for our purposes, they also point out that the type of constraint is “less influential than typically assumed” (1). Below, we argue that increasing novel constraints introduces novel dimensions on which an artist's work might vary, thereby introducing new ways in which chance might constrain a work. Such “chancy constraints” introduce novel sources of creativity for the artist. Our proposal is consistent with existing cognitive science literature on the relationship between constraints and creativity. Other models of creativity that also emphasize the importance of constraints include: Stokes (2014) constraint-based model and Tromp and Sternberg's Integrated Constraints in Creativity (IConIC) model (Tromp, 2023; Tromp & Sternberg, 2022). While the positive role of constraints for creativity has been demonstrated experimentally, our goal is to further substantiate this view using two case studies (see e.g. Medeiros et al., 2014, 2018). What we hope to contribute to this literature is a particular emphasis on how chance acts as a type of constraint in dynamical systems wherein creativity is being enacted. The two cases we focus on were not chosen to be representative of a particular group of artistic practices, but rather as outliers, particularly strong examples of how chance acts as a productive constraint in artistic creativity. To see how our view differs from

core contributions within the established literature, consider how these accounts treat creativity as a computational process:

In one of the most influential works on the cognitive science of creativity, Boden (2004) distinguishes three forms of creativity. The first involves generating unfamiliar combinations of familiar ideas as when one thinks of poetic imagery or a new melody; the second is the exploration of conceptual spaces, as is done by scientists and philosophers; the third is realizing the limitations of conceptual spaces and transforming them to surpass those limits. For present purposes, the key uniting factor in these three types of creativity is that they all occur in the realm of ideas, presumably within the brains of creative thinkers. Importantly, Boden's view builds on orthodox views in cognitive science and treats creativity, like all cognitive processes, as essentially computational. Other authors have already emphasized the importance of constraints for creativity, but likewise do so within a representationalist framework. For example, Johnson-Laird (2002) gives a central place to constraints as one of the five elements of his "NONCE" account of creativity, but the constraints he is talking about are musical and esthetic, and in keeping with his computationalist perspective, the way they constrain the creative process is as representations in the individual's mind (p. 420). On this view, the way in which musicians "work within the constraints of a genre and of their own personal style" when improvising involves having "an internal representation of the musical criteria governing the generation of melodies" (Johnson-Laird, 2002, pp. 419, 422). Similarly, Haught and Johnson-Laird (2003) claim that "constraints are at the heart of the creative process," but the constraints in question are all inside of people's heads (p. 532): Haught and Johnson-Laird describe how the subjects retrieve "knowledge about the entities in question" and "use this knowledge as a constraint," where "the constraints [are] embodied in the representation of these entities" (p. 532). In

contrast, we are interested not in the constraints inside people's heads, but in the constraints that emerge from the dynamic interactions between humans and their material environments. Against the orthodoxy that this literature builds on, we argue that the representationalist paradigm in cognitive science is mistaken and intend this paper as a contribution to a non-representationalist account of creativity.

Tim Ingold offers a sophisticated account and critique of the view of creativity we are arguing against under the label of the "hylomorphic model." He argues that "contemporary discussions of art and technology, and of what it means to make things, continue to reproduce the underlying assumptions of the hylomorphic model" (Ingold, 2010, p. 92). The hylomorphic model is not the original Aristotelian account of how form and matter are related, but a variant of it that has developed throughout the history of Western thought into the version that is widespread today and is exemplified in views like Boden's. According to the hylomorphic model, creativity is the generation of novel ideas and creative activity is the imposition of these ideas onto matter. In contrast to this, Ingold proposes a view of creativity in which creativity is in the act of making, "the tactile and sensuous knowledge of line and surface," as opposed to a more abstract stance in which form is comprehended through geometry first and only then related to a material substrate (Ingold, 2010, p. 92). This shift from an active, haptic engagement with the world toward detached contemplation importantly involves a shift of focus from a practical to a theoretical image of the world. It is in this sense that Ingold's aim, to "overthrow" the hylomorphic view and "replace it with an ontology that assigns primacy to the processes of formation as against their final products" echoes and amplifies the critical stance of enactive cognitive science toward the cognitivist orthodoxy (Ingold, 2010, p. 92).

What Ingold criticizes as an ontology derived from the historical dominance of the "technical" over the "textilic" corresponds closely to

the flawed account of creativity we are trying to replace, in that both are based in the conviction that “things are constituted in the rational and rule-governed transposition of preconceived form onto inert substance” (Ingold, 2010, p. 93). In contrast to this, Ingold sees humans making things as “weaving their own lines of becoming into the texture of material flows comprising the lifeworld” (Ingold, 2010, p. 96). Ingold provides vivid accounts of architecture and cooking to illustrate how different practices of making involve “not so much imposing form on matter as bringing together diverse materials and combining or redirecting their flow in the anticipation of what might emerge” (Ingold, 2010, p. 94).

In what follows, we supplement these examples with an analysis of the same basic principle within the domains of musical improvisation and conceptual art. One point on which we can add to Ingold’s view concerns his rejection of the tendency “to read creativity ‘backwards’, starting from an outcome in the form of a novel object and tracing it, through a sequence of antecedent conditions, to an unprecedented idea in the mind of an agent” (Ingold, 2010, p. 97). Once we understand the creative process as involving a maker enmeshed in a dynamic system of material processes and interrelations (Glăveanu, 2015; Ross & Vallée-Tourangeau, 2021; Ross et al., to appear), complexity theory can show why this sort of backwards inference fails: Sufficiently complex systems, such as those exhibiting self-organization, develop in ways that cannot be deduced from their starting conditions according to pre-given laws. Instead, once they reach a point of bifurcation, we cannot predict which of several new trajectories they will embark on. Complexity introduces the arrow of time into a reversible picture of Newtonian physics and throws a wrench in simple schemas of sequential inference. Instead of reading backwards from a material result in search for an original intention, Ingold invites us to read “creativity ‘forwards’,” with “a focus not on abduction but on *improvisation*” (2010, p. 97).

Complexity, constraint, and improvisation

There is a plausible scientific story to be told about the relationship between constraints and creativity in improvised musical performance based in complex systems theory. The core assumptions for this scientific story involve *self-organization*. A self-organizing system is a system that exhibits regularities that arise without a plan or leader but emerge from the interactions of the parts of the system. The posit that human action is self-organizing began the recent resurgence of interest in dynamical systems models in psychology (Kugler et al., 1980). Dynamical systems models work by assuming that thinking, experiencing, acting humans are self-organizing systems that comprise portions of their brains, bodies and tools. Self-organizing systems have their organization without a plan or controller. Consider, to take a mundane example, the whirlpool pattern that forms when a toilet is flushed. The water molecules temporarily move in a coordinated way that is very atypical for water molecules. They can exhibit this pattern of behavior because potential energy from the water in the tank is released when the toilet is flushed. This energy enables the pattern to form and last until the energy is dissipated, after which the water molecules go back to their more typical behavior. Physicists who work on self-organization call this behavior of water molecules in a flushing toilet a *synergy* (Haken, 2007; Kelso, 2009). A synergy is a collection of components that act as a temporary self-organizing unit, and in which the components exhibit behavior that is constrained by being parts of the synergy. See Figure 1, Riley et al. (2011). The top portion of the figure depicts a set of components as separate from each other, for example a collection of water molecules in an unflushed toilet. The bottom portion depicts the synergy: constraints acting on the already constrained networks of molecules so that a coherent collective pattern of activity (a whirlpool) emerges.

Synergies might seem odd from a mechanistic point of view: a synergy is an aggregate

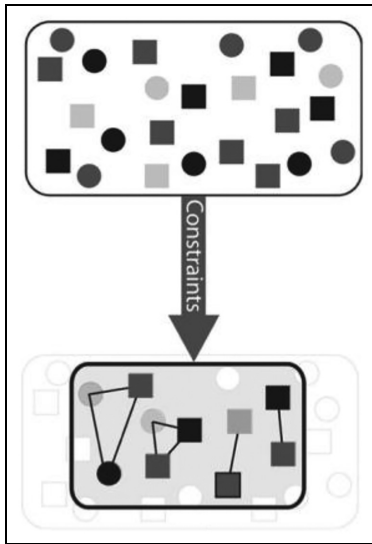


Figure 1. Synergy.

Note. Courtesy of Michael Riley.

entity that constrains the behavior of the smaller-scale components that make it up, implying a kind of downward or circular causation that strains our ordinary understanding of causes and effects. Nonetheless, there is nothing magical about synergies—they occur routinely in nature and are well understood by physicists. The pattern in your flushing toilet is a synergy, as are convection rolls and bird flocks. In each, components form leaderless, temporary coalitions in which the behavior of the components is constrained by being part of the whole that they are parts of. Many well-functioning physiological systems are synergies. Examples of this include human heartbeats (Peng et al., 1995) and gait patterns (Hausdorff et al., 1995), indicating that the chambers of the heart, the locomotory system, and parts of the human brain are synergies. More recently, synergies have made their way into the cognitive sciences. For example, there is mounting evidence that synergies are the chief mechanism according to which brain areas work together to enable and control behavior—brain areas move into temporary coordinated patterns in order to enable animals to complete a task and become uncoordinated

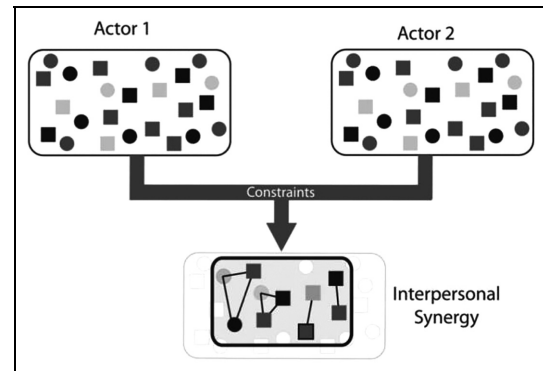


Figure 2. Interpersonal synergy.

Note. Courtesy of Michael Riley.

once the task is completed (Anderson, 2014; Bressler & Kelso, 2001; Freeman, 2006).

Notice that synergies occur at multiple scales, including in groups of organisms. For example, in bird flocks, no particular bird is in charge of the speed or direction of the flock, and the behavior of individual birds is determined by the behavior of the flock that they make up. This is also common in human interpersonal behavior. When you dance or jog with someone, you and your partner form a temporary unit, in which each of you allow your behavior to be constrained by the pair you comprise. The earliest experimental work on interpersonal synergies was done by Schmidt et al. (1990). They showed that while two people who can see one another swing their limbs in synchrony, they are connected to one another in the same way that the limbs of a single individual are connected. That is, the interpersonal connection between two people is a temporary version of the intrapersonal connection among the parts of a single person. Riley et al. (2011) called this *interpersonal synergy*. See Figure 2. More recent work has shown similar interpersonal synergies arising whenever people engage in a cooperative task; such a task constrains them to alter their behavior so that they can work together (e.g. Abney et al., 2014; Abney et al., 2021). Here we will focus on a set of experiments on jazz improvisation.

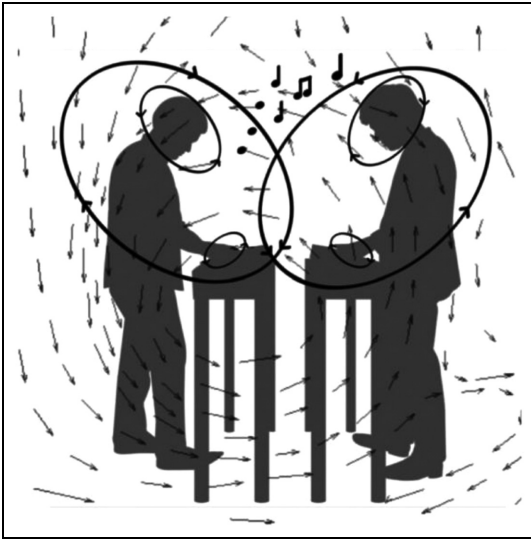


Figure 3. Experimental setup.
Note. Courtesy of Ashley Walton.

Walton et al. (2015), Walton, Washburn, Langland-Hassan et al. (2018), Walton et al. (2018) studied the synergies that arise in jazz performance. She brought professional jazz piano players into the lab in pairs, suited them up in motion capture sensors, and had them improvise together over different backing tracks: a single-tone drone, an ostinato (a short, repeating musical phrase consisting of four ascending chords), and the chord progression from a jazz standard. These backing tracks represent increasing levels of task constraint provided by the backing track; each performer's playing provided further constraints. She also had the musicians perform either in sight of one another or with a curtain drawn. She measured the musicians' hand movements and head movements, both up and down and side to side. See Figure 3 for the experimental set up. At the beginning of the experiment, the musicians first performed two individual warm-up trials, where they improvised once over each backing track while their coparticipant waited outside the lab. Then together the pairs performed two sets of four 2-min improvised duets for a total of eight duets. Afterward, the musicians were

interviewed separately about the performance while listening to their final few duets.

Analyzing the movement data of the players, Walton and her colleagues found that the musicians participate in an interpersonal synergy, specifically by tightly coordinating their head movements, nodding along with the backing track and with one another. Walton and her colleagues did not find similar coordination in the players' hand movements. While the movements of the hands may be considered more relevant for keyboard playing, velocity of head movements has been found to play a large role in performance expressivity (Castellano et al., 2008; Juchniewicz, 2008). This head-movement synergy is the foundation for their melodic improvisation; the interpersonal synergy is the set of constraints that enables that improvisation. The melodic creativity of the performance arises in the context of, and even because of, the constraints imposed by the backing track and the players' tightly coordinated head movements. We can take this scientific study of collaborative improvisation as an example of what Ingold calls "morphogenesis," since it illustrates central aspects of how novelty, including novel constraints, emerges from the spontaneous coordination of interactions, rather than from the imposition of a pre-existing intention on a material substrate. Instead of the notion of creativity as an achievement of individuals that proceeds from a mental spark to a physical form, the pattern that constrains the play of both musicians emerges from the process that unfolds between them.

Constraint, composition, and conceptual art

Two case studies from the world of art can highlight in more detail the productive role that constraints play in for creative expression. Conversely, artists often challenge and sometimes even displace existing constraints, creating situations in which chance and improvisation can lead to the emergence of novel constraints which in turn afford new creative

trajectories. We illustrate these principles in the interplay between composition and improvisation in the work of the composer Anthony Braxton and in Marcel Duchamp's interventions at the intersection of sculptural and conceptual art.

The compositions of Anthony Braxton

We typically think of musical composition and improvisation as diametrically opposed. Composition dictates in advance the form a piece of music will take by imposing constraints via musical notation, while improvisation frees a musician by removing those constraints. Of course, in practice, it's not that simple. Subtle interpretations must be given in order to play any composition. No matter how many constraints are imposed, each time a piece of music is played, there will be slight variations. And most improvisations occur against the backdrop of music that has been composed, typically with one instrumentalist improvising while the others play the score. And when two or more musicians improvise, regardless of whether or not there are any compositional constraints, each musician's choices place constraints on the other(s). Even in cases where a single musician improvises without any compositional constraint, self-imposed constraints emerge as patterns are noticed and repeated (or avoided), largely determined by one's esthetic proclivities and general disposition in the moment. Thus, a more nuanced look at the relationship between composition and improvisation reveals that musical constraints cannot prevent variability across performances. The difference between composition and improvisation is not that there are constraints in the former but not in the latter, instead improvisational constraints merely increase the range in which performances may vary. And by increasing that range of variability, they allow for the morphogenetic emergence of creativity through the interplay between performers and improvisational constraints. Given this, composition and improvisation are best understood on a continuum.

As a case study, experimental musician Anthony Braxton exploits this relationship between musical constraints and improvisational creativity. Instead of limiting the constraints he imposed on himself and his collaborators, he increased them. His compositions typically included geometric shapes and colors intended to constrain the mood, tone, rhythm, or timbre of the performers, sometimes with, and sometimes without, traditional musical notation. By adding such constraints, Braxton's compositions provide performers with more dimensions on which their improvisations might vary. In other words, the additional constraints introduced additional degrees of freedom. Such constraints invite interpretation, thereby expanding, rather than limiting, the space for creative expression. Moreover, these constraints play a critical role in Braxton's multi-instrumental improvisations.

For example, Steinbeck (2018) discusses Braxton's *Composition 76*, his first piece to include what he refers to as "modular notation." The score consists of "forty-one unique 'modules' written on twenty cards that can be arranged in any order," where each module consists of graphical syntax indicating both composed and improvised passages (p. 254). It is written for three (multi-instrumental) performers. The traditional five-line staff is employed but in fragmentary pieces, with combinations of numerical notation (indicating how many notes to play in an improvisational section), colored shapes (indicating improvisational bursts), various symbols (indicating when performers should switch instruments or use their voice), and various other codes (indicating when a performer should "dominate," "support," or be "open" in a way that is neither dominant nor supportive with respect to the other performers).

By providing the performers with additional dimensions on which the improvisation may vary, these idiosyncratic constraints introduce many different domains in which chance may influence the music. In other words, some of the variance across the additional dimensions

on which the improvisation may vary will be due to chance. Such stochasticity acts as an additional source of constraint on the performers. As mentioned above, responding to the other performers' improvisation acts itself as a constraint on each performer. And because much of the variance in the performers' improvisation will be due to chance, we can see that chance itself introduces an additional type of constraint. These novel sources of constraint provided by chance thus afford the performers with novel sources of creativity.

Steinbeck describes the general attitude of the Association for the Advancement of Creative Musicians (AACM) (of which Braxton was a core member) toward composition and improvisation as follows:

"Crucially, the association's compositional turn did not lead to a wholesale adoption of concert-music ideology. AACM members were especially reluctant to perpetuate the outmoded divide between composers and performers, inventors and interpreters. Such distinctions made little sense to Braxton and his AACM colleagues, who played all of these roles at a single concert. With every piece they composed, AACM members provided one another with platforms for improvisation. These improvisations, in turn, yielded musical ideas that became new compositions. Accordingly, the AACM began to view composition and improvisation as two sides of the same coin, two facets of a creative practice that was collaborative at its core. This, according to Braxton (1988, p. 4:143), is why he "integrate[d] color and shape variables into the operational scheme of [Composition 76]": he wanted "to generate fresh creative responses from [the] instrumentalists," who could create—collectively and spontaneously—a work richer than what he could have constructed on his own" (p. 258).

What we see (and hear) from Braxton and company is a rejection of a hylomorphic conception of both composition and improvisation. What we hear when listening to a Braxton composition is not the prescribed work of a sole composer that has given complete structure to a piece of music by translating some finalized, ideal

product in his mind into a notated reality. Moreover, it is also not the sound of a disembodied creative mind lacking material constraints to both a written piece of music and other performers bringing some ideal product into existence in real time. Rather, Braxton's music demonstrates the range and depth of creativity to be found by embedding additional constraints across multiple dimensions of our material existence, including written musical form and one's relation to other performers (by being, e.g. dominant, supportive, or open), one's instrument (by, e.g. switching instruments, or playing the same instrument in a specific, non-traditional manner), and one's own body (by, e.g. vocalizing in non-traditional ways).

Marcel Duchamp's conceptual sculptures

Marcel Duchamp was one of many artists who ushered in a new way of thinking and making art in the early 20th century. His work *The Fountain* is now recognized as one of the most influential works of art ever suggested, suggested because the production of this work was technically at a factory that produced urinals in the United States. See Figure 4. In fact, this work of art was created by pushing on the constraints of what and what could not be an art object in the milieu of America in 1917. The main constraint was that art (here sculpture) had to be made by hand. Photography had already eroded this constraint in painting, but sculpture was another matter. Duchamp's "fountain," which in its original incarnation was a store bought urinal flipped over and signed "R. Mutt," would come to exemplify how far "conceptual art" would push against the constraints of several definitions of "art" and establish a new way of making art that is still the norm more than 100 years later. Precursors to the 1917 *Fountain*, a series of artistic episodes Duchamp called "readymades," all attempt to reconfigure the constraints that allowed certain objects to be thought of as art. The aftermath of this series



Figure 4. Duchamp's (1917) *Fountain* (public domain).

of works changed the way art was thought of and made going forward. Duchamp's new paradigm suggested that art no longer fit into the previous constraints of painting, sculpture, etc. but instead was determined by a new set of criteria, or constraints if you will, that were established then and are today lumped together as conceptual art practice.

Duchamp's work culminating in *The Fountain* shows how the whole enterprise of conceptual art depends upon changing the constraints on both what art is and what it can be. Furthermore, the method of Duchamp and his allies was not simply to push against the constraints of what is and is not art in his time, but to set up the perpetually changing constraints necessary in order to judge something like a urinal as art. Just as it was impossible to judge a urinal as a sculpture under the old regime of what people considered sculpture to be, one needs to be given new constraints, a new imaginary vocabulary, to think of a urinal as art. Thus, one had to change the constraints that allowed for sculpture to be and not to be a

urinal in addition to pushing against the original constraints of sculpture. Thus, the method of conceptual art is to both push against existing constraints and introduce new ones, continuously challenging and changing the constraints on art making that make possible the creative process. These ideas can be illustrated in more detail by looking at one of Duchamp's pre-urinal works, *The Three Standard Stoppages* of 1913-14. (See also Ross et al, to appear.)

Years before *the Fountain*, when Duchamp was thinking about what makes art "art," he was creating a series of works, or "experiential set-ups" that demonstrated his focus on using constraints or art domains to create so-called "anti-art." Anti-art, or "readymades," are a series of objects that play with the constraints of sculpture, painting, etc. *The Three Standard Stoppages* of 1913-14 reveal the experimental set up of Duchamp's idea and the methodology for creating conceptual art works that simultaneously depend upon constraints and change them, only to then start the cycle again in what we might call "creative (epi)cycles." Particularly, for the three stoppages piece, Duchamp was interested in exploring the role of chance as art inquiry, in opposition to the logic found in scientific experimental set-ups. Thus "chance" was the first constraint Duchamp used in making this piece: "Pure chance interested [him] as a way of going against logical reality" (Duchamp, quoted in Molderings, 2010, p. 2). The three stoppages piece is an example of how Duchamp's artistic practice sets up "chance" as an action in art via a set of constraints. Duchamp set up constraints to make the work happen. First he set up the materials, which consisted of three black canvas strips. He then "dropped" three separate white threads themselves cut at 1 m onto the black canvas. Once the threads had fallen onto the canvas, he then attached the threads on the canvases how they had fallen (according to him). This sets up the "new standard" meter consisting of 1 m long "fallen" threads, but curved by chance, interrupting the standard of 1 m used in measurement.

Here Duchamp used chance to create a set of new constraints to proceed from. His new meter would measure the world in a new way, and constrain it in a new way, too. Following these initial thread drops, he produced tread curves from wood to be used as new “standard meters.” These were to be used as draftable edges to replicate his new “meter.” One can imagine creating further pieces using Duchamp’s new “standard meter,” or how the world would appear if measured against it. Of course this all is in some sense ridiculous, but Duchamp, who was deliberately echoing the platinum-iridium bar in Paris, which is also in a nice wooden box, was seeking to point out that the original meter was also connected to the creation of constraints for scientific measurement, being one-ten millionth the distance from the equator to the north pole. How is the human creation of constraints different in the two cases? The metric system’s constraints produced an object, the meter stick. This object became a standard for measurement, requiring, as in Duchamp’s thread version, a permanent physical reference to disseminate and replicate the unit, and to then produce new work in science. In much the same way, Duchamp’s the *Three Stoppages* provided the standard meter for art.

In his mocking mimicry of the scientific method, Duchamp also highlighted aspects of the scientific process as born of constraints. Duchamp’s role as a conceptual artist should be understood in light of his view that scientific methods should be brought into the fold of artistic inquiry with the intent of producing art works from constraints—even if the constraints are imaginary, or culturally constructed. As Duchamp scholar Molderings states: “Duchamp did not acknowledge science as the prime source of human knowledge, but developed an art form that undermined science’s claim to reveal the laws of nature and life, countering it with imaginative processes of poetic and pictorial thought. ‘Art,’ Duchamp concluded toward the end of his life, ‘is the only thing left for people who don’t give science the

last word” (Molderings, 2010, p. 144). Duchamp saw science, like art, as an outgrowth of something deeper, human inquiry. While this inquiry required material constraints, for Duchamp the constraints of thinking and culture were even more important.

This is evident in the play of matter in the meter work, where he doesn’t really care about the material of “physics of chance,” but how we *think* they are functioning. In fact, upon a deeper look, the *Three Stoppages* was not even created by the constraints Duchamp himself selected. Given the way the treads are presented, it is obvious he did not drop the threads, but arranged them esthetically by hand in order to create a “painterly” aspect to the back canvas with white treads. The white threads are not actually 1 m long either, and given that the whole work was worked and reworked several times, the physical object being produced, remixed, and reproduced to fit thinking and the cultural forms of conceptual art at different times, we should conclude that the stated physical constraints are not as important as the idea of constraints for creating the work.

To summarize the three main points: First, Duchamp wants us to see the connection between the standard meter in Paris and the construction of the wooden case and tools which both reference and complicate it. This emphasizes how humans set standards through the material production of physical constraints, which are then replicated. The metric system returns to material constraints, which are created from cultural constraints, which then constrain how we imagine distance, weight, and mass, etc. Secondly, Duchamp wants us to understand that the scientific method, or “repeatable experience” as science, is itself a “set-up.” Scientific experiments, like the three falling threads in artistic inquiry and like the performance of a musical score, can yield different results when repeated. This is a comment on the supposed repeatability of science via experimental parameters (constraints) that yield results similar enough to generalize knowledge. Scientific logic or rigor is a “set-up” just as

much as Duchamp's art work on chance. Third he reveals that, on closer inspection and reflection, his artistic "experiments" can be considered successful despite never actually being performed according to the specified method. Unlike in science, having staged his results for maximum visual effect does not invalidate the results of his artistic inquiry. By intentionally placing the threads to create an esthetic arrangement, he demonstrated that the human subjective constraint is always present and productive. As a whole, the *Three Stoppages* project addressed the role of cultural and human constraints in art and scientific inquiry.

The Fountain was created a few years after the *Three Standard Stoppages* was first executed. It seems that Duchamp moved from an artistic investigation of the role that constraints play in setting the standards of (re)production in art and science toward an intervention in the cultural constraints on what may count as art in the first place. Duchamp's work highlights how material and cultural constraints interact in ways that are themselves mutually constraining. The urinal can only be considered art according to the changed cultural constraints on conceptual and sculptural art that Duchamp himself brought about with this very piece, *The Fountain*. As urinal, it is subject to cultural constraints, while as *Fountain*, it disregards, disrupts, and finally determines the cultural constraints on what counts as art from then on. Similarly, Anthony Braxton's work crosses the boundaries between different sensory modalities and shows how composition and improvisation, seemingly paradigmatic of fixed rigidity and spontaneous fluidity, are not logical opposites but two intricately connected aspects of a process enabling the emergence, repetition, and recording of novelty.

Conclusion

Taking our cue from Tim Ingold, we have sketched out some case studies in order to argue against the "hylomorphic" view of creativity and in favor of a "morphogenetic"

account. If creativity were a matter of novelty emerging first in the mind and being imposed as form on a material substrate in a second step, constraints would feature only as obstacles in the creative process. In contrast, and in reality, creativity is not an individual achievement that originates from some hidden mental realms, but the result of complex processes of interaction between different numbers of agents and varying sets of material, cultural, and social constraints.

Our case studies provide confirmation for this morphogenetic account of creativity by highlighting the role of constraints as enabling the emergence of novelty. This novelty itself sometimes takes the form of new constraints, which in turn afford new creative trajectories to be explored by human agents, or by chance. The jazz pianists studied by Ashley Walton and her colleagues improvise in a way that is constrained both by the backing tracks provided in the experimental set-up and by the dynamic constraints emerging from their own interaction (2015, 2018a, 2018b). Anthony Braxton's "modular notation" enables subtly textured creativity at the interface between composition and improvisation along the multiple dimensions of musicians' engagement with each other, their instruments, and their bodies. The groundbreaking artistic interventions by Marcel Duchamp changed the very cultural constraints on what counts as sculpture, played a key role in the inauguration of conceptual art as a creative genre, and critically interrogated the relationship between chance and constraints for the generation of novelty and replicability in art and science.

One common thread that runs through our case studies is the role of chance in mediating the interplay between constraints and creativity. Jazz musicians and conceptual artists alike navigate spaces of possibilities afforded by specific situations of constrained creativity. Exploring these spaces, we might understand them as "enacting chance" in a way that in turn leads to the emergence of novel creative constraints. Just as novelty in the arts depends on the constraints

that exist in a given situation and emerge from its development over time, the conceptual vocabularies provided (or prohibited) by different scholarly disciplines provide creative constraints for the generation of new insights and perspectives. Our own approach to the study of complex social phenomena such as creativity is informed in particular by the confluence of radical embodied cognitive science, arts-based research, and some thinkers from the dissident fringes of anthropology, archeology, and related disciplines (cf. Feiten et al., 2021, 2022). We believe that great potential lies in exploring the intersections between art and science in new ways, beyond the confines of accepted disciplinary norms, to challenge existing constraints on scholarship and enable the emergence of new ones.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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Note

1. We would like to express our sincere gratitude to four anonymous reviewers whose comments significantly improved this paper.

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